INNOVATIVE APPLICATIONS OF INDUSTRIAL ROBOTICS

Robotic Cell for Automated Helmet Visor Testing and Visual Inspection

Authors

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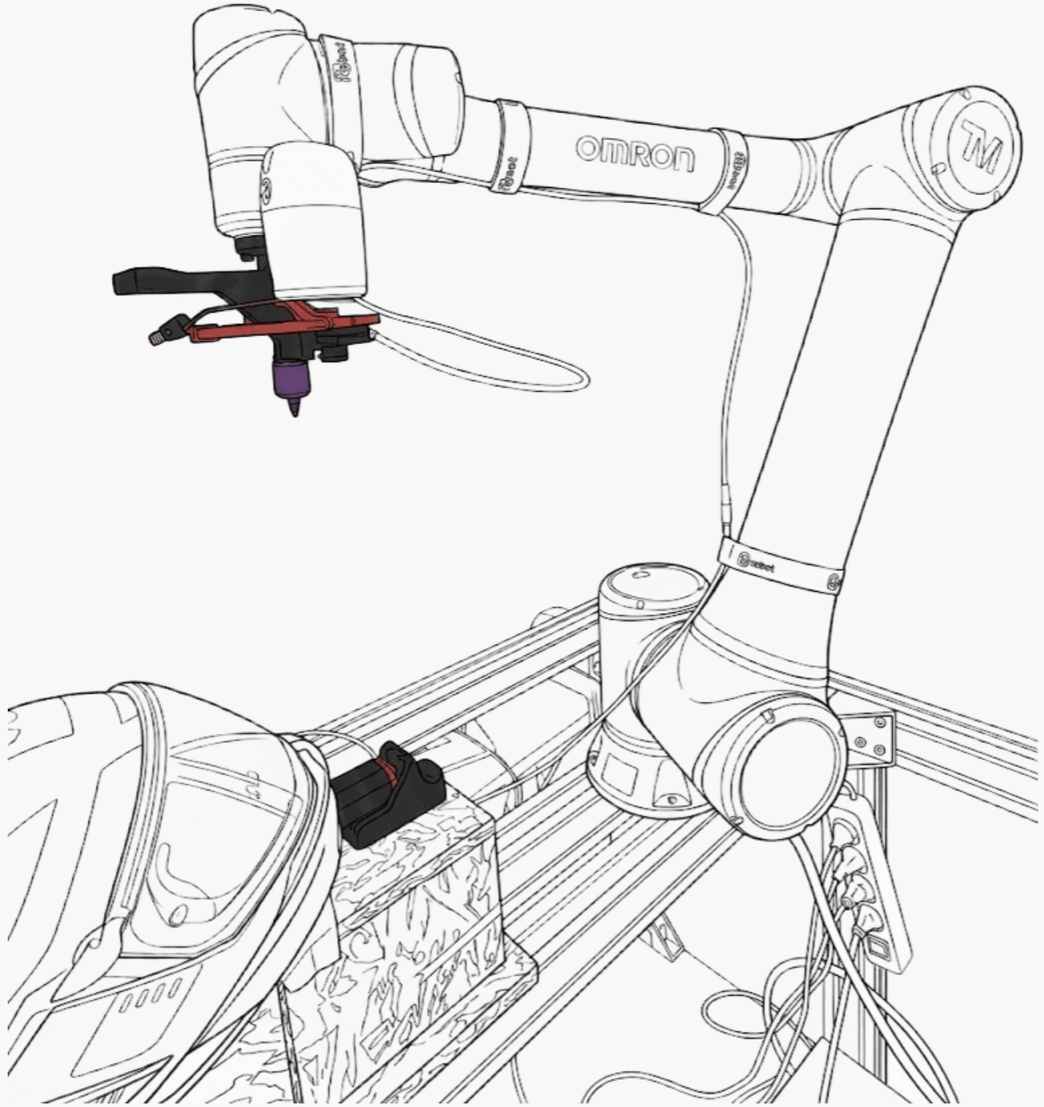
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Supervisor

Giovanni Bianchi

Academic Year

2025 / 2026



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THE GOAL

PROJECT OBJECTIVES

Six pillars — each one anchors a section of this deck

01

CONSTRAINING

Constrain the helmet rigidly & repeatably

02

END EFFECTOR

Integrated tool: handling, vision & marking

03

HANDLING

Robot control & Visor tasks

04

ACQUISITION

Capture the surface with the stereo-camera

05

DETECTION

Locate surface defects automatically

06

INSPECTION & MARKING

Identify, refine and mark each detected defect

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SECTION 01 / 06

Constraining

Constrain the helmet rigidly and repeatably

01 CONSTRAINING

02 END EFFECTOR

03 HANDLING

04 ACQUISITION

05 DETECTION

06 INSPECTION & MARKING

THE OBJECT

ANATOMY OF THE HELMET

Key features the cell must handle



01 **Visor tab**

Tab that help for the opening of the visor

02 **Sun-visor levers**

Slider that opens the inner sun visor and button to close it

03 **Chin bar release button**

A mechanism to release the lower part of the helmet

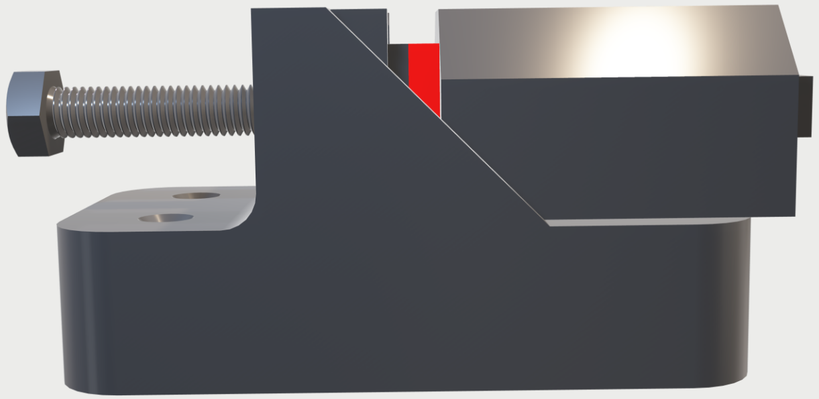
04 **Inspection zone**

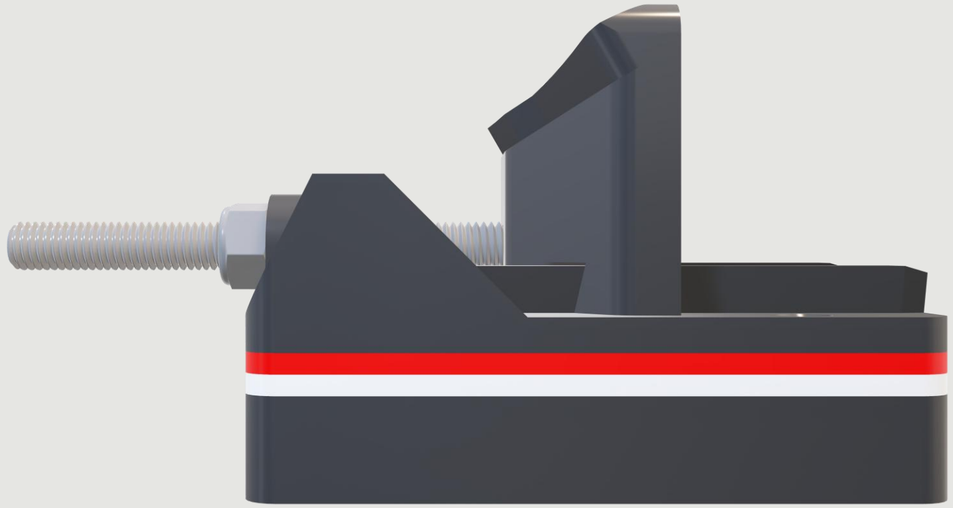
Zone of the helmet we have to inspect to find eventual damage

THE DESIGN

DESIGNING THE FIXTURE

Modular, tool-free and repeatable by design





01
STABILITY
Rigid under external loads

02
REPEATABILITY
Same position every cycle

03
ACCESSIBILITY
Clear approach for visors, marking & to complete the taks

04
MODULAR
Adjustable - fits all helmet sizes

05
FAST-MOUNT
Mounts & dismounts rapidly and easily

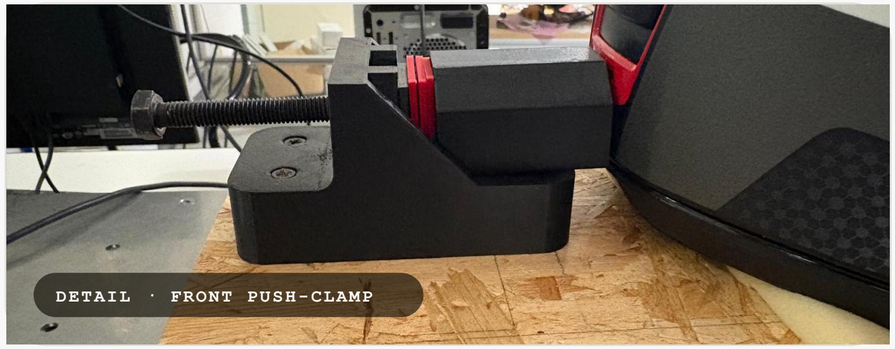
IMPLEMENTATION

THE CLAMPING SYSTEM

Realised fixture — front push-clamp and rear support base



PHOTO · HELMET CONSTRAINED IN THE CELL



DETAIL · FRONT PUSH-CLAMP



DETAIL · REAR SUPPORT BASE

- ***Key idea: the helmet remain in the exact same position on every cycle.***

FORCES





Reaction Forces:
Applied Forces:

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SECTION 02 / 06

End Effector

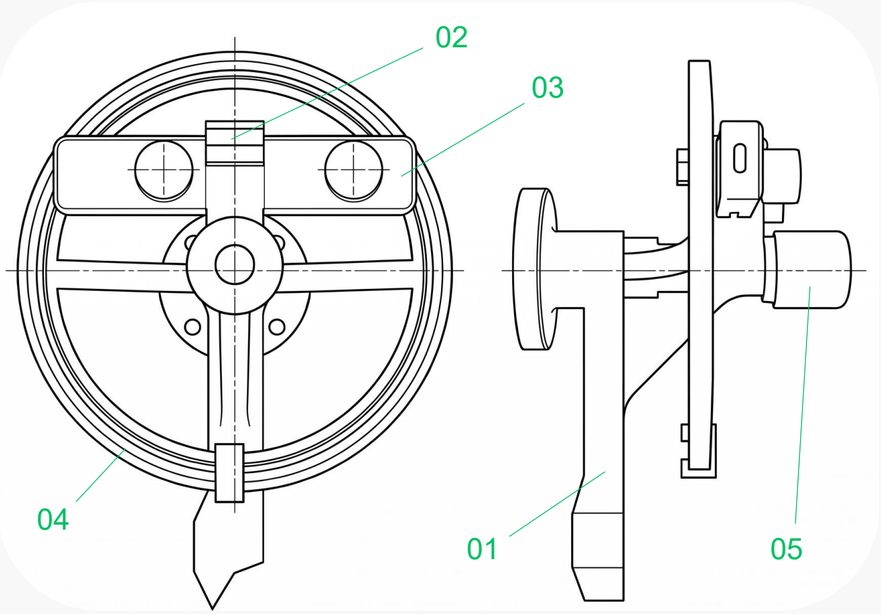
One integrated tool for handling, vision and marking

01 CONSTRAINING
02 END EFFECTOR
03 HANDLING
04 ACQUISITION
05 DETECTION
06 INSPECTION & MARKING

THE DESIGN

END-EFFECTOR DESIGN

CAD model and bill of materials



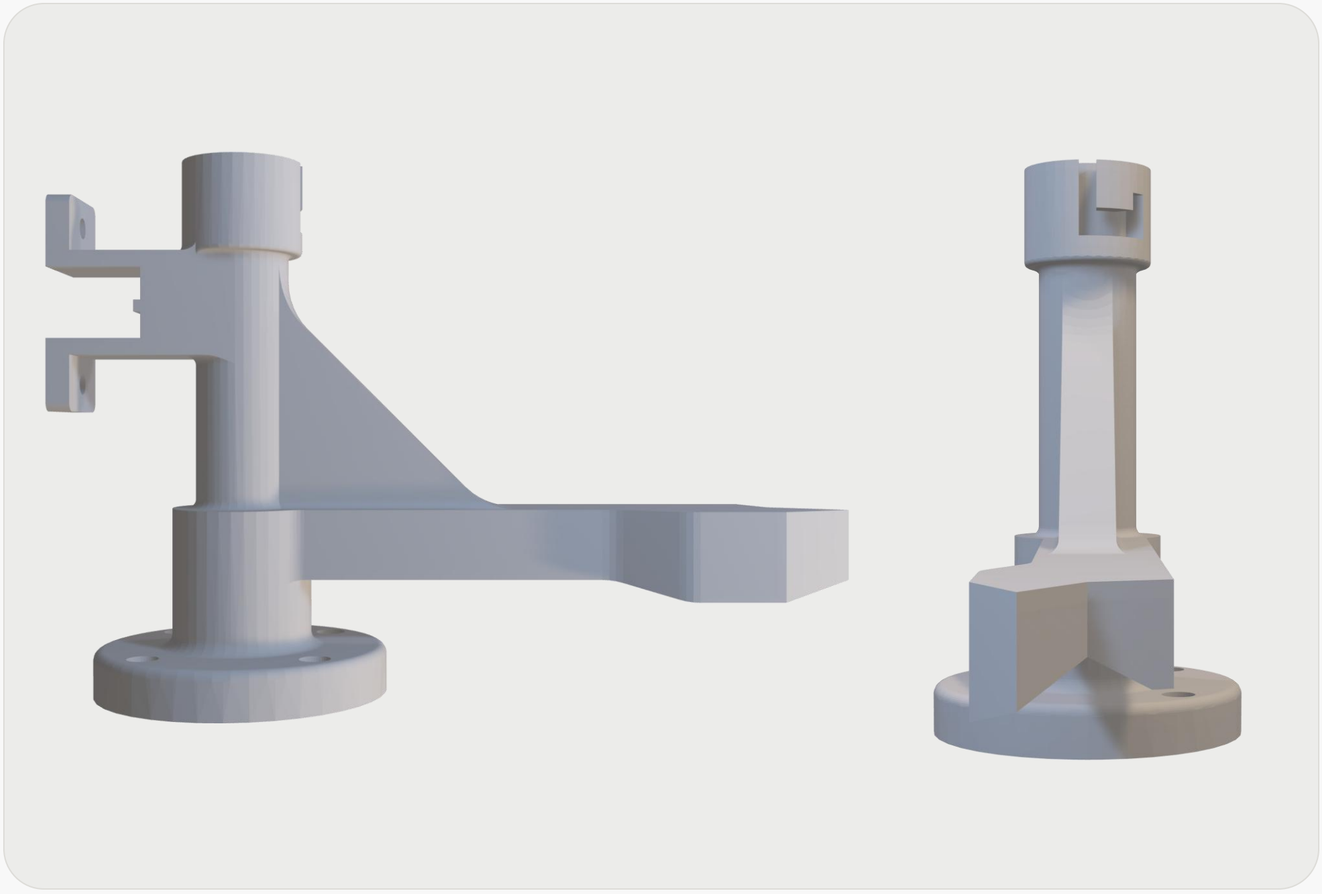
BILL OF MATERIALS

#	COMPONENT	QTY
01	**Base + Integrated Finger** Rigid frame, robot mount & finger — One printed piece	1
02	**Stereo-camera support** Holds the ZED mini	1
03	**ZED mini stereo-camera** 3D surface acquisition	1
04	**LED ring + support** Illumination system	1
05	**Marker system** Cap, Marker, Spring – Marking defects	1

IMPLEMENTATION

ASSEMBLING THE END-EFFECTOR

Progressive build-up — one component at a time



STEP 1 / 5 · COMPONENT ADDED

BASE + INTEGRATED FINGER

FUNCTION

Structural backbone of the entire end-effector, perform manual tasks and house camera and marker

RATIONALE

3D-printed design with optimized layer orientation

ADVANTAGE

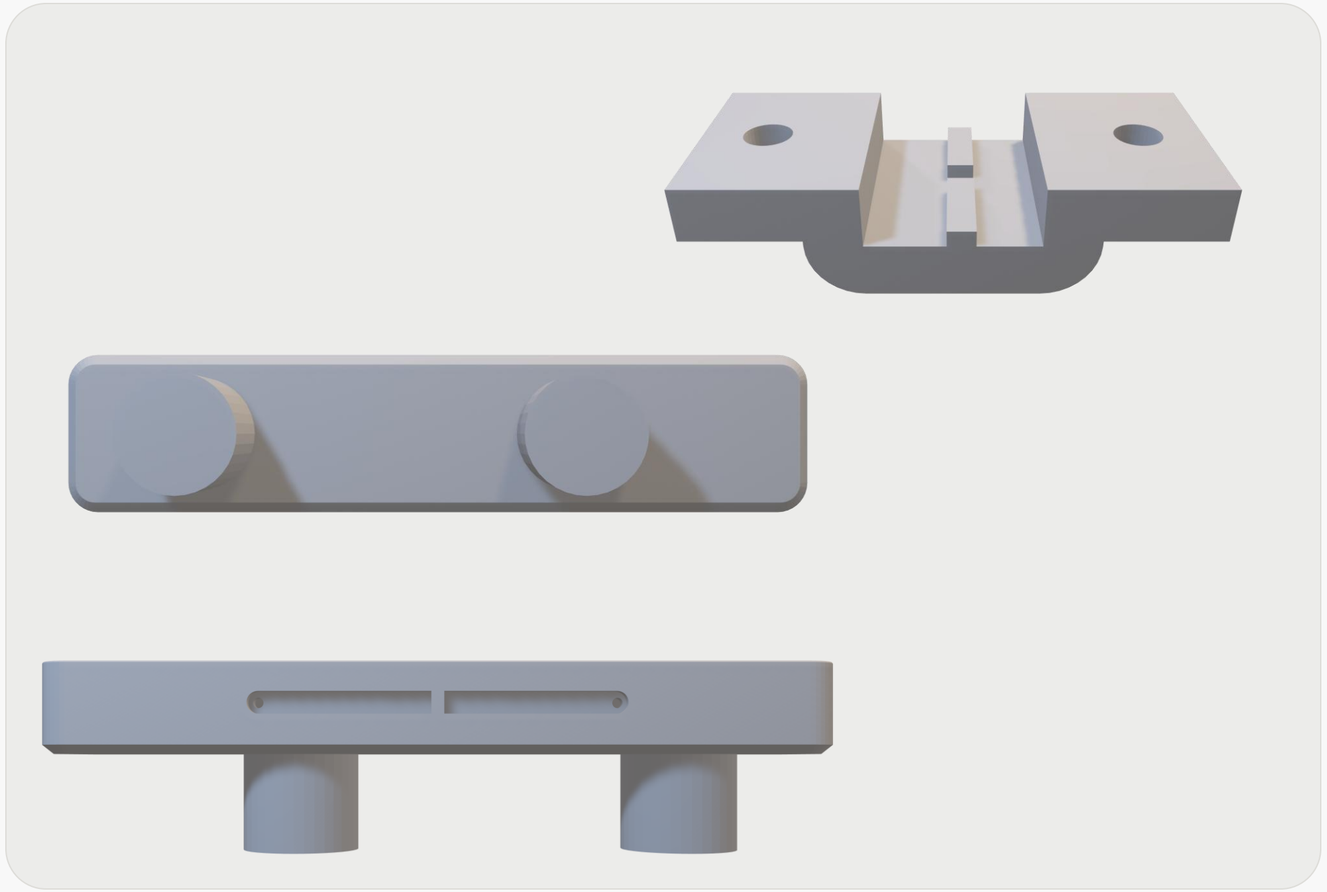
Maximum strength — no joint or fastener to fail

- **Base + integrated finger**
- Stereo-camera support
- LED support
- Marker system

IMPLEMENTATION

ASSEMBLING THE END-EFFECTOR

Progressive build-up — one component at a time



STEP 2 / 5 · COMPONENT ADDED

STEREO-CAMERA SUPPORT

FUNCTION

Holds the ZED mini that captures the 3D surface

RATIONALE

Custom matching grooves designed to interlock perfectly with the camera's original housing

ADVANTAGE

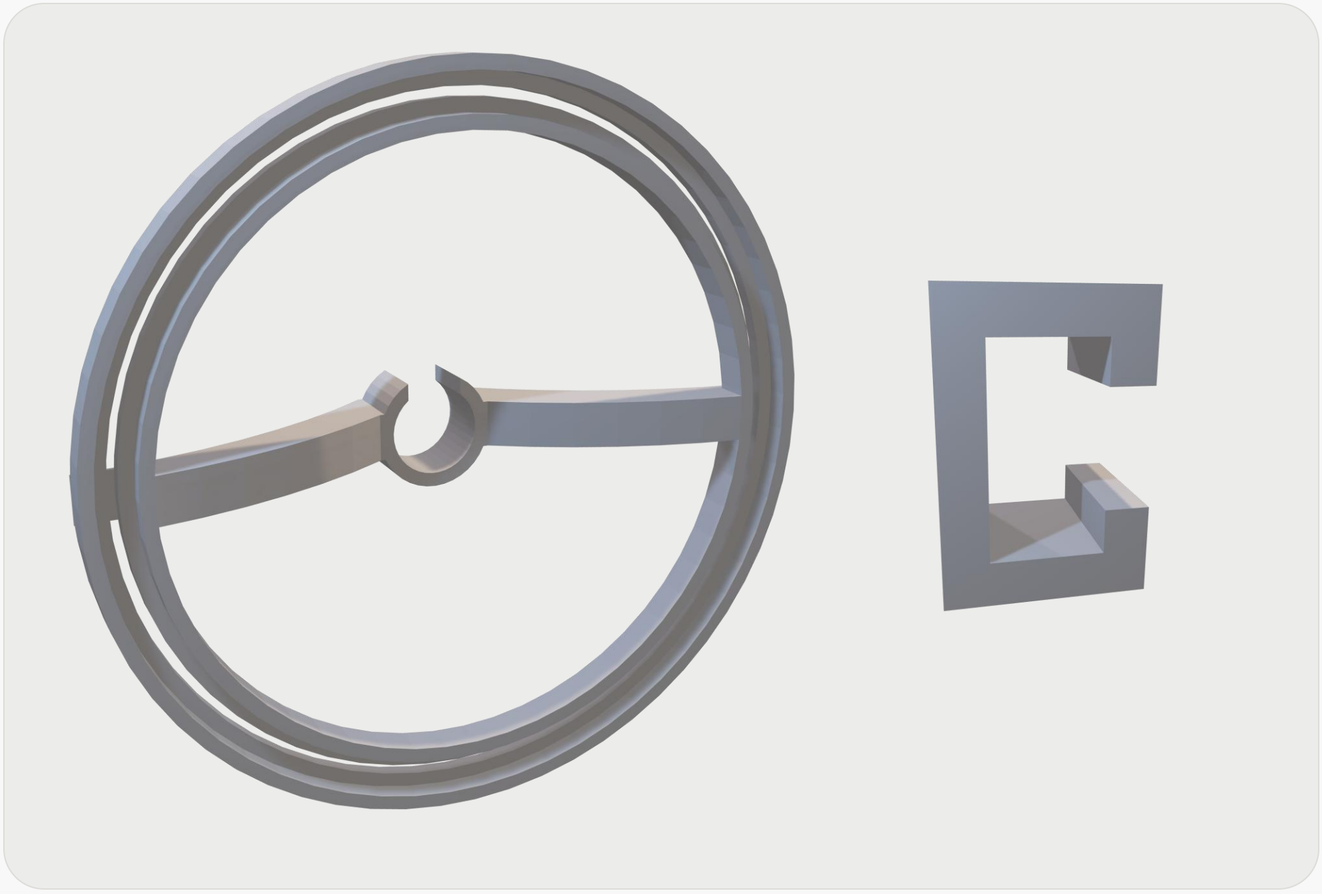
Self-align assembly, easy to disassemble

- **Base + integrated finger**
- **Stereo-camera support**
- LED support
- Marker system

IMPLEMENTATION

ASSEMBLING THE END-EFFECTOR

Progressive build-up — one component at a time



STEP 3 / 5 · COMPONENT ADDED

LED SUPPORT

FUNCTION

Mounts the LED ring for uniform illumination

RATIONALE

Snap-fit mechanical locking

ADVANTAGE

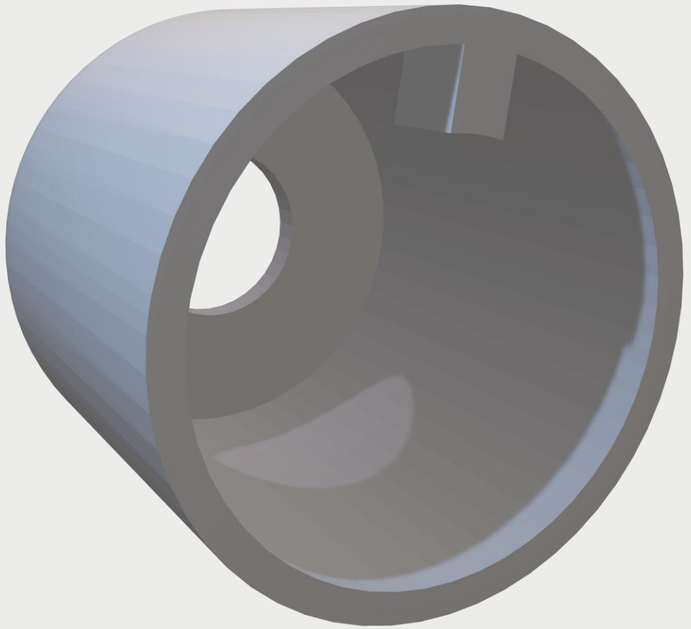
Easy to disassemble, no impact on the robot's workspace

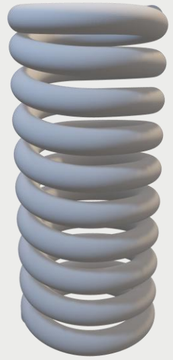
- **Base + integrated finger**
- **Stereo-camera support**
- **LED support**
- Marker system

IMPLEMENTATION

ASSEMBLING THE END-EFFECTOR

Progressive build-up — one component at a time





STEP 4 / 5 · COMPONENT ADDED

MARKER SYSTEM

FUNCTION

Secures the marking pen and dampens contact forces

RATIONALE

Axial spring-loaded compliance, locking system with main body

ADVANTAGE

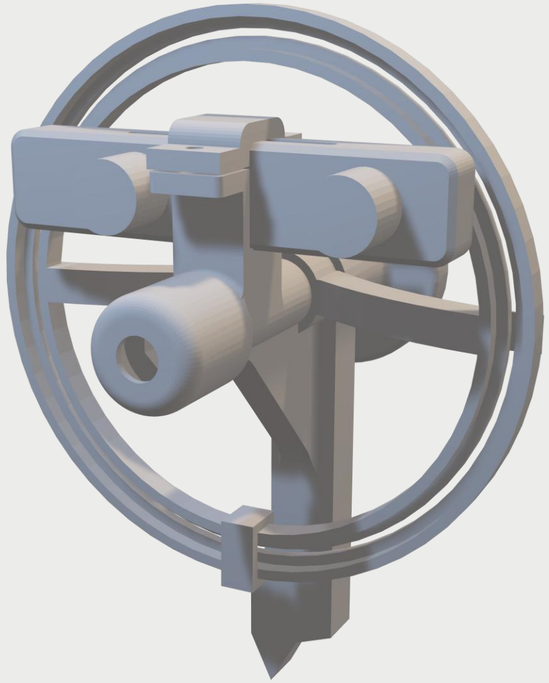
Retractable pen, Easy to mount and dismount

- **Base + integrated finger**
- **Stereo-camera support**
- **LED support**
- **Marker system**

IMPLEMENTATION

ASSEMBLING THE END-EFFECTOR

Progressive build-up — one component at a time





STEP 5 / 5 · COMPLETE

COMPLETE ASSEMBLY

FUNCTION

Handling, vision, lighting & marking in one tool

RATIONALE

One end-effector for the entire cycle

ADVANTAGE

No tool changes — faster inspection

- **Base + integrated finger**
- **Stereo-camera support**
- **LED support**
- **Marker system**

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SECTION 03 / 06

Handling

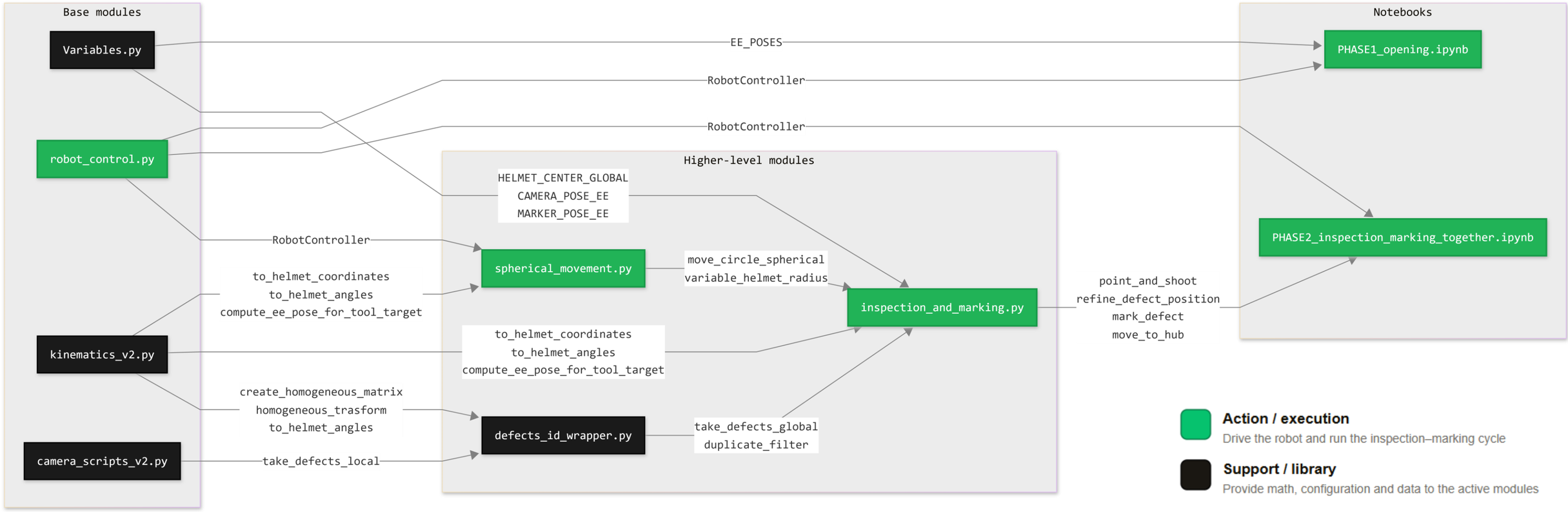
Robot control & Visor tasks

01 CONSTRAINING
02 END EFFECTOR
03 HANDLING
04 ACQUISITION
05 DETECTION
06 INSPECTION & MARKING

HANDLING

SOFTWARE ARCHITECTURE

Modular architecture, orchestrated by two workflows



HANDLING

ROBOT CONTROL

Listen Node commands, Modbus TCP feedback



01 **COMMAND INTERFACE**

A single class sends motion scripts to the Techman Listen Node and reads the pose back over Modbus TCP.

02 **SYNCHRONOUS & BLOCKING**

The program waits until the robot reaches the target within a tolerance, by continuously polling the pose

03 **MOTION PRIMITIVES**

Joint moves, Cartesian point-to-point, linear moves on the TCP, and 3D arcs

04 **SAFETY**

Home pose, emergency stop and a movement timeout

HANDLING

REFERENCE FRAMES & THE MATRIX H

One 4×4 carries rotation and translation

$$H = \begin{bmatrix} r_{11} & r_{12} & r_{13} & t_x \\ r_{21} & r_{22} & r_{23} & t_y \\ r_{31} & r_{32} & r_{33} & t_z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

R T

$$P_{(global)} = H \cdot P_{(local)}$$

01 **FOUR REFERENCE FRAMES**

Robot base - End-Effector (TCP) - tool (camera and marker, each with its own fixed offset) - helmet

02 **ONE MATRIX H (4×4)**

Rotation **R (3×3)** + translation **T (3×1)** in a single block takes a point from local to global

03 **DIRECT & INVERSE**

The direct chain $H_{(cam \to global)} = H_{(ee \to global)} \cdot H_{(cam \to ee)}$ brings a camera-space defect into the robot base frame.

The inverse H^{-1} does the opposite, global → local, we use it to strip the tool offset and recover the EE pose: $H_{ee} = H(_{target)} \cdot H^{-1}_{(tool \to ee)}$

HANDLING

ROTATION MATRIX ↔ ANGLES

Z-Y-X (Cardan) convention, aligned to the Techman controller

CONSTRUCTION

$$R = R_z(r_z) \cdot R_y(r_y) \cdot R_x(r_x)$$

Z-Y-X intrinsic ≡ X-Y-Z extrinsic — matches the robot controller's Euler convention

EXTRACTION

$$s_y = \sqrt{({r_{11}}^2 + {r_{21}}^2)}$$

$$\mathrm{r}_x = atan2\,(\,r_{32}, r_{33}\,)$$

$$\mathrm{r}_y = atan2\,(\,-r_{31}, s_y\,)$$

$$\mathrm{r}_z = atan2\,(\,r_{21}, r_{11}\,)$$

$$R = \begin{bmatrix} \cos(r_z)\cos(r_y) & \cos(r_z)\sin(r_y)\sin(r_x) - \sin(r_z)\cos(r_x) & \cos(r_z)\sin(r_y)\cos(r_x) + \sin(r_z)\sin(r_x) \\ \sin(r_z)\cos(r_y) & \sin(r_z)\sin(r_y)\sin(r_x) + \cos(r_z)\cos(r_x) & \sin(r_z)\sin(r_y)\cos(r_x) - \cos(r_z)\sin(r_x) \\ -\sin(r_y) & \cos(r_y)\sin(r_x) & \cos(r_y)\cos(r_x) \end{bmatrix}$$

01 ATAN2, ALL FOUR QUADRANTS

Resolves the angle with the correct sign and avoids division by zero

02 SINGULARITIES

When sy ≈ 0 is pushed into zones the robot cannot reach

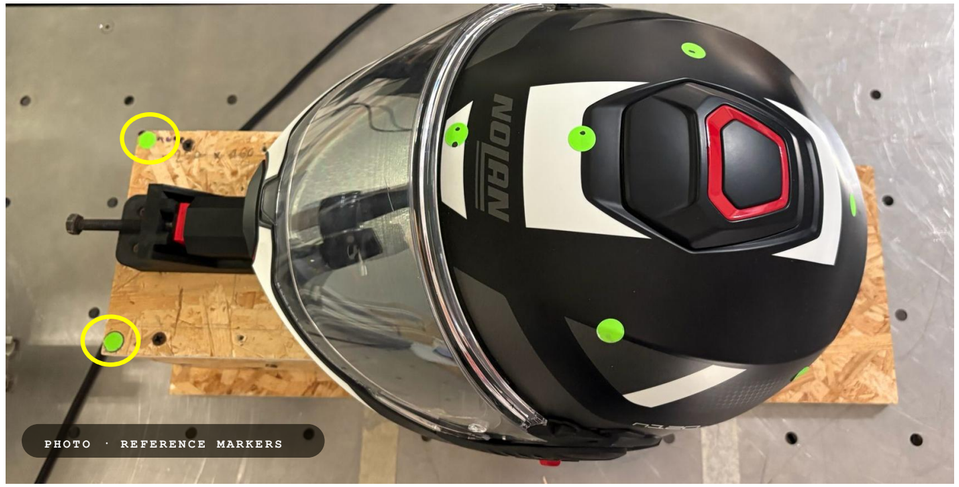
HANDLING

FROM OUR ROBOT TO ANY ROBOT

Generalising the poses so the cell is Adaptable



Omron TM12





Omron TM5

01

ABSOLUTE POSES

Hand-taught poses, valid only for that cell and that robot, not universal

02

MARKER-FRAME POSES

Each taught pose is re-expressed relative to a fixed reference frame on the base fixture (a marker) as:

$$\widetilde{P_{loc}} = H_{base}^{-1}\,\widetilde{P_{taught}}$$

03

RE-PROJECTION

Given the new marker pose, every pose is mapped back to the robot base as:

$$\widetilde{\mathrm{P_{new}}} = \mathrm{H_{new}}\,\mathrm{H_{base}^{-1}}\,\widetilde{\mathrm{P_{taught}}}$$

04

ADAPTABILITY

For a new cell or robot, just re-measure the marker pose. The whole opening/closing sequence re-projects automatically

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SECTION 04 / 06

Acquisition

Capture the surface with the stereo-camera

01 CONSTRAINING
02 END EFFECTOR
03 HANDLING
04 ACQUISITION
05 DETECTION
06 INSPECTION & MARKING

ACQUISITION

AUTOMATED HELMET INSPECTION SYSTEM

3D Computer Vision & Robotic Kinematics for Defect Control

01 **HELMET CENTER**

The core coordinate

02 **STEREO-CAMERA**

Hardware configuration: Eye-in-Hand-system (camera rigidly attached to the EE)

03 **3D SURFACE SCANNING**

Continuos, automated 3D surface scanning ensuring perpendicularity

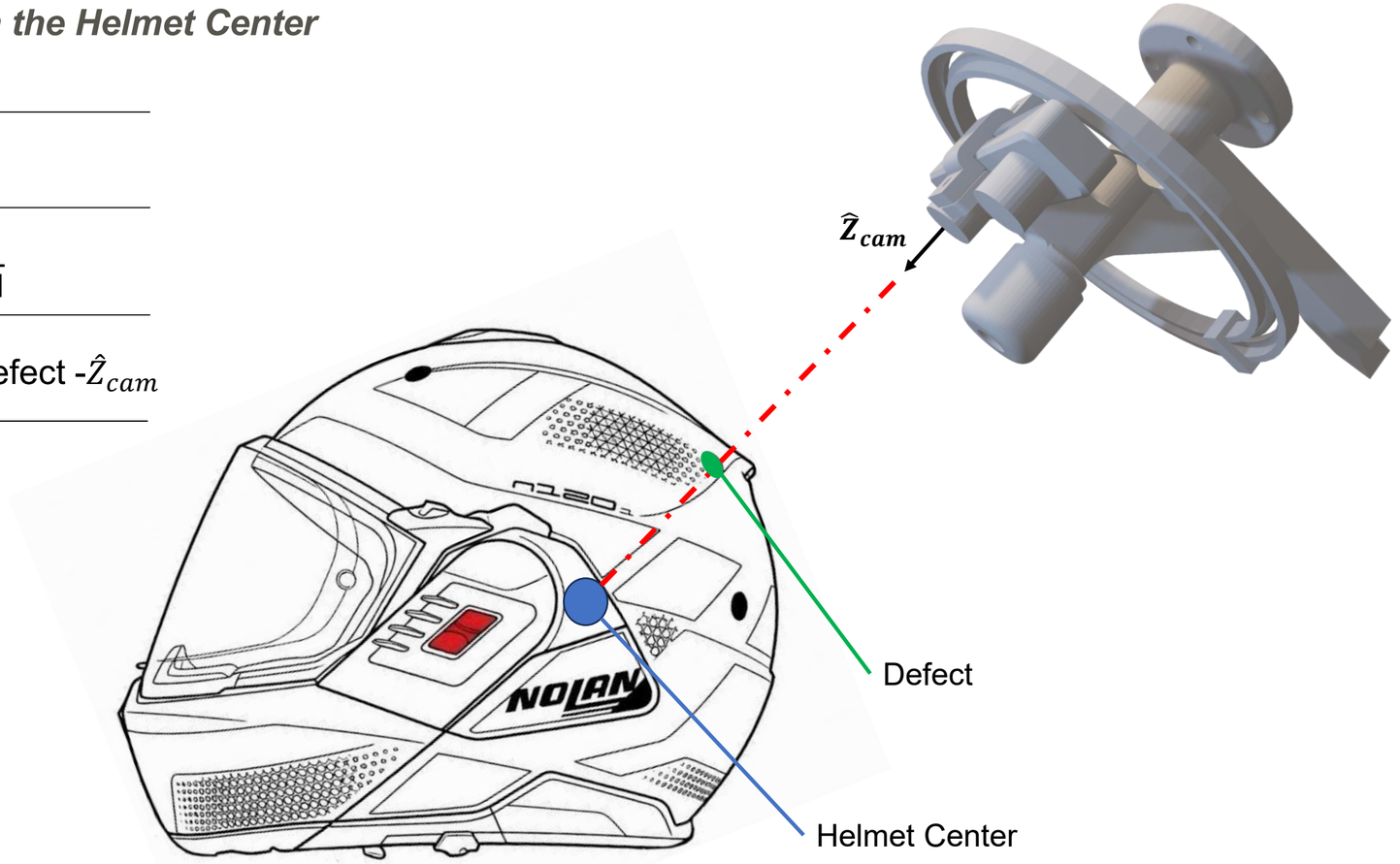


ACQUISITION

ORIENTATION CONSTRAINT: LOCAL Z-AXIS TRACKING

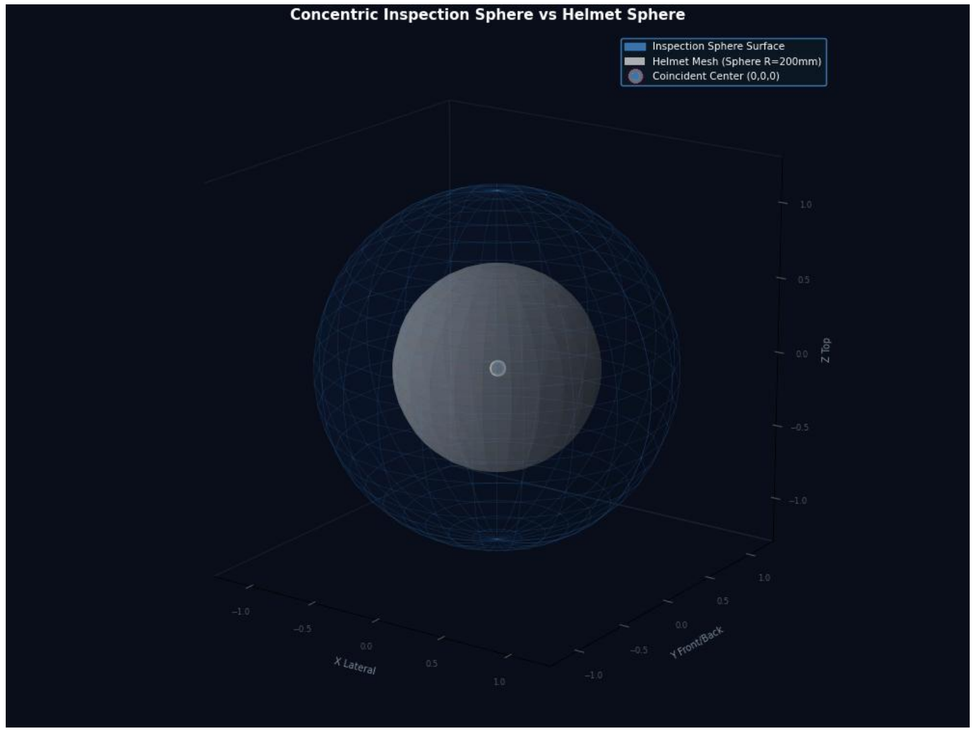
Ensuring Constant Optical Perpendicularity on the Helmet Center

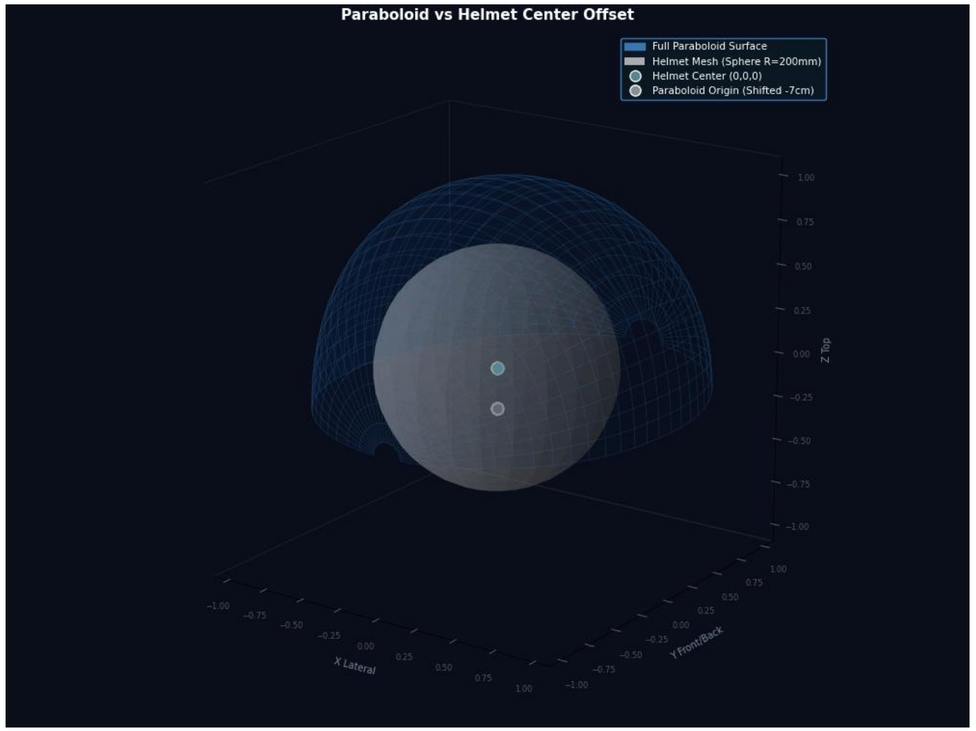
EE POSE	Defined by $p = [\, x \,,\, y \,,\, z \,]^T$
Z_{cam} ALIGNMENT	Camera Versor : $\hat{Z}_{cam} = \frac{-p}{\|p\|}$
ADVANTAGE	Alignment Helmet Center – Defect -$\hat{Z}_{cam}$



ACQUISITION

TRAJECTORY GENERATION: VARIABLE HELMET RADIUS (VHR)





01

THE PROBLEM

Standard spheres trigger upper-shell collisions or miss lower helmet collar

02

SOLUTION

Paraboloid flaring shape that opens up as elevation decreases

03

DESIGN RADIUS

$R_{APEX} = 380\text{mm}$

$R_{SIDE/BACK} = 300\text{mm}$

$R_{FRONT} = 450\text{mm}$

04

TRIGONOMETRIC TRANSITION

Based on angle α (azimuth) and β (elevation)

ACQUISITION

COLLISION AVOIDANCE & VIRTUAL ENVIROMENT VALIDATION

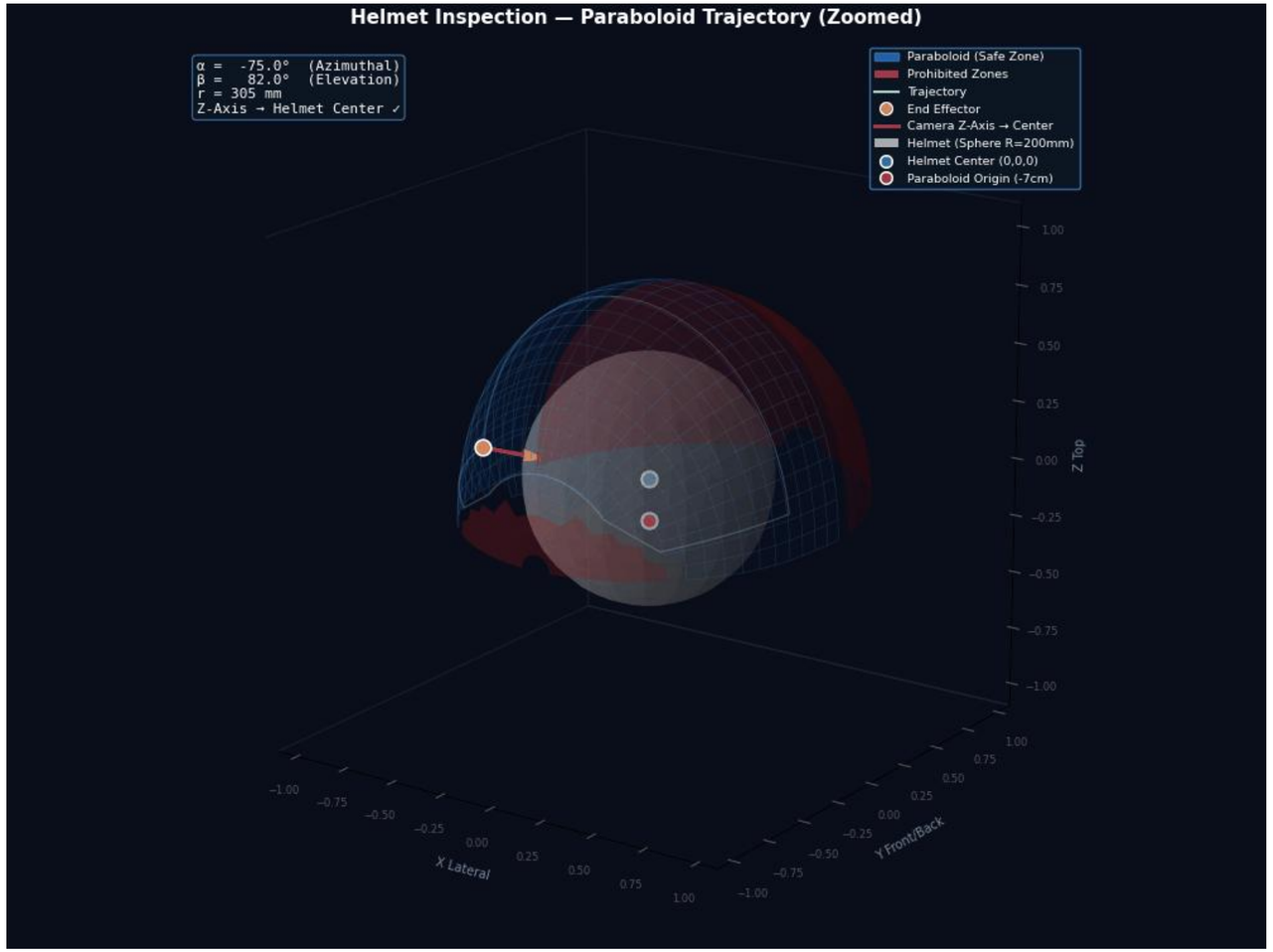
Bridging Algirithmic Safety with Digital Twin Validation

UNSAFE ZONES

01	**Gimbal Lock Prevention**	$\lvert\alpha\rvert > 89°$
02	**Low Boundary Collision**	$\beta < 20 + 25\left(\frac{\alpha}{90}\right)^2$
03	**Out-of-Bounds Limit**	$\beta > 110°$

DIGITAL TWIN PARAMETERS

01	**Helmet modeled as a sphere**	$R = 200mm$
02	**Paraboloid centre**	$[0, 0, -70]$
03	**Scaling Normalization**	$1/R_{APEX}$



ACQUISITION

IMAGE ACQUISITION

Stereo-camera capture images during the scan



RGB gives the 2D evidence; XYZ gives the depth information.

01

RGB IMAGES

Used for colour masks and 2D defect extraction

02

XYZ POINT CLOUD

Used to estimate the defect position in camera frame

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SECTION 05 / 06

Detection

Locate the surface defects automatically

01 CONSTRAINING
02 END EFFECTOR
03 HANDLING
04 ACQUISITION
05 DETECTION
06 INSPECTION & MARKING

DETECTION

2D DEFECT DETECTION: FROM MASK TO CENTROID

HSV segmentation is cleaned and converted into isolated defect candidates

01 **MASK GENERATION**

Direct green mask

Green known colour

BGR → HSV → green threshold

Inverse anomaly mask

For generic unknown colour

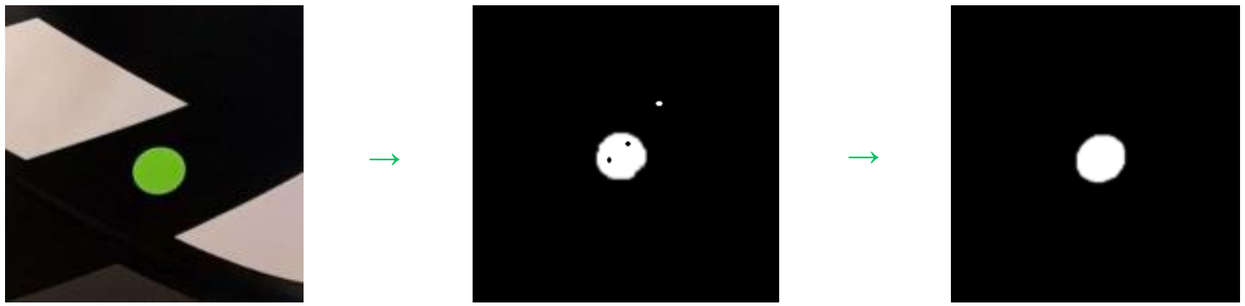
BGR →HSV → helmet colours threshold → inversion of the mask

02 **MASK CLEANING**

Attention radius →useful field of view

Opening → removes isolated noises

Closing → fills small holes



DETECTION

2D DEFECT DETECTION: FROM MASK TO CENTROID

HSV segmentation is cleaned and converted into isolated defect candidates

03 **BLOB FILTERING**

Find external contourns

Reject small blobs by area threshold

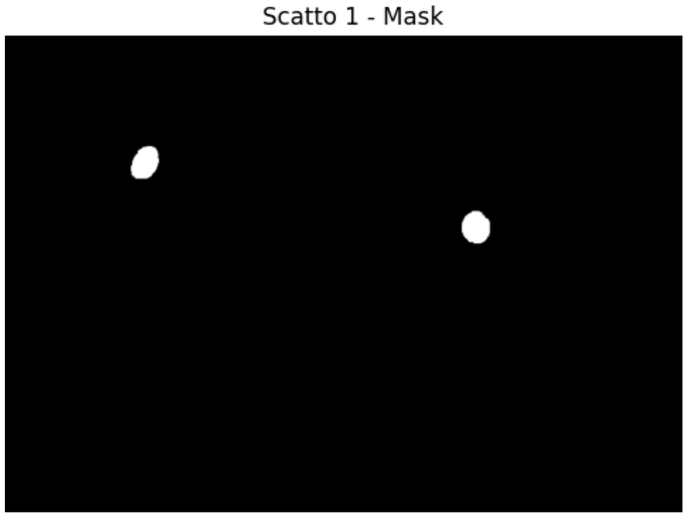
04 **CENTROID & ISOLATED MASK**

Compute area and centroid from image

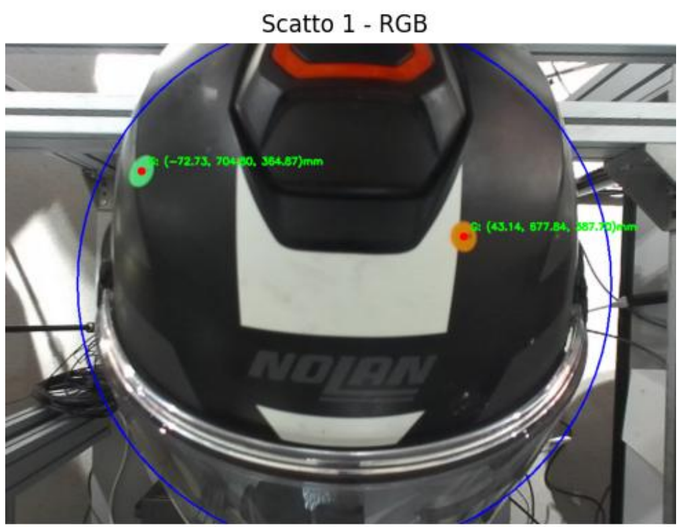
Store one isolated mask for defect



→



→



DETECTION

FROM 2D MASK TO 3D DEFECT POSITION

White mask pixels are projected into the ZED point cloud and averaged

01 **POINT CLOUD LOOKUP**

From white pixels of isolated defect mask

Pixel (x,y)→ [Xc,Yc,Zc] in camera frame

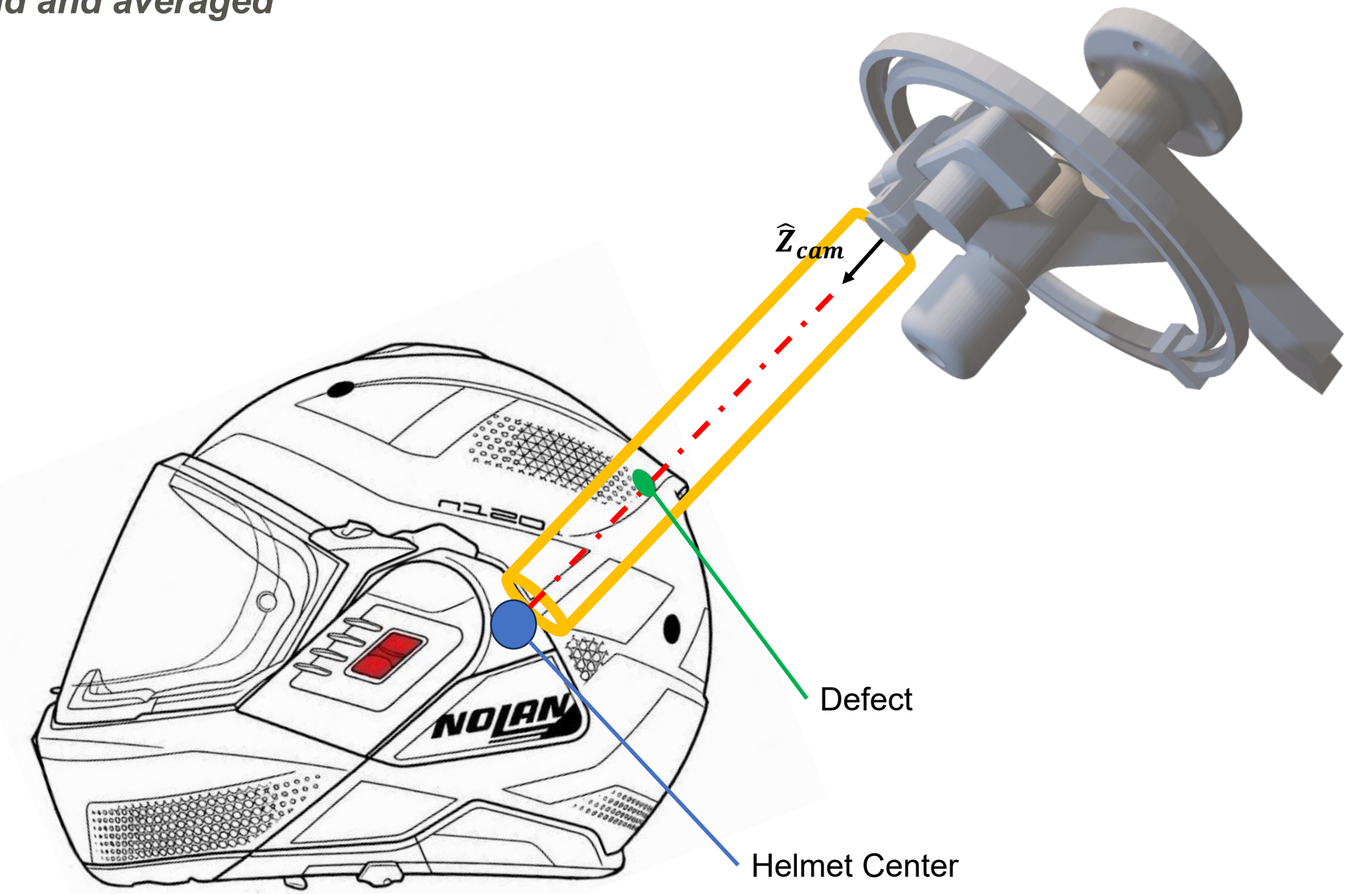
02 **VALID XYZ SAMPLES AND CYLINDER FILTER**

Reject failed,NaN or invalid depth values

Apply filter (lateral radius and depth range)

03 **MEAN 3D POINT**

Mean of 3D points of one defect in camera frame



DETECTION

FROM CAMERA FRAME TO HELMET TARGET

The 3D defect points are transformed from camera frame to the robot global frame and then expressed relative to the helmet

01 CAMERA-TO-GLOBAL TRANSFORM

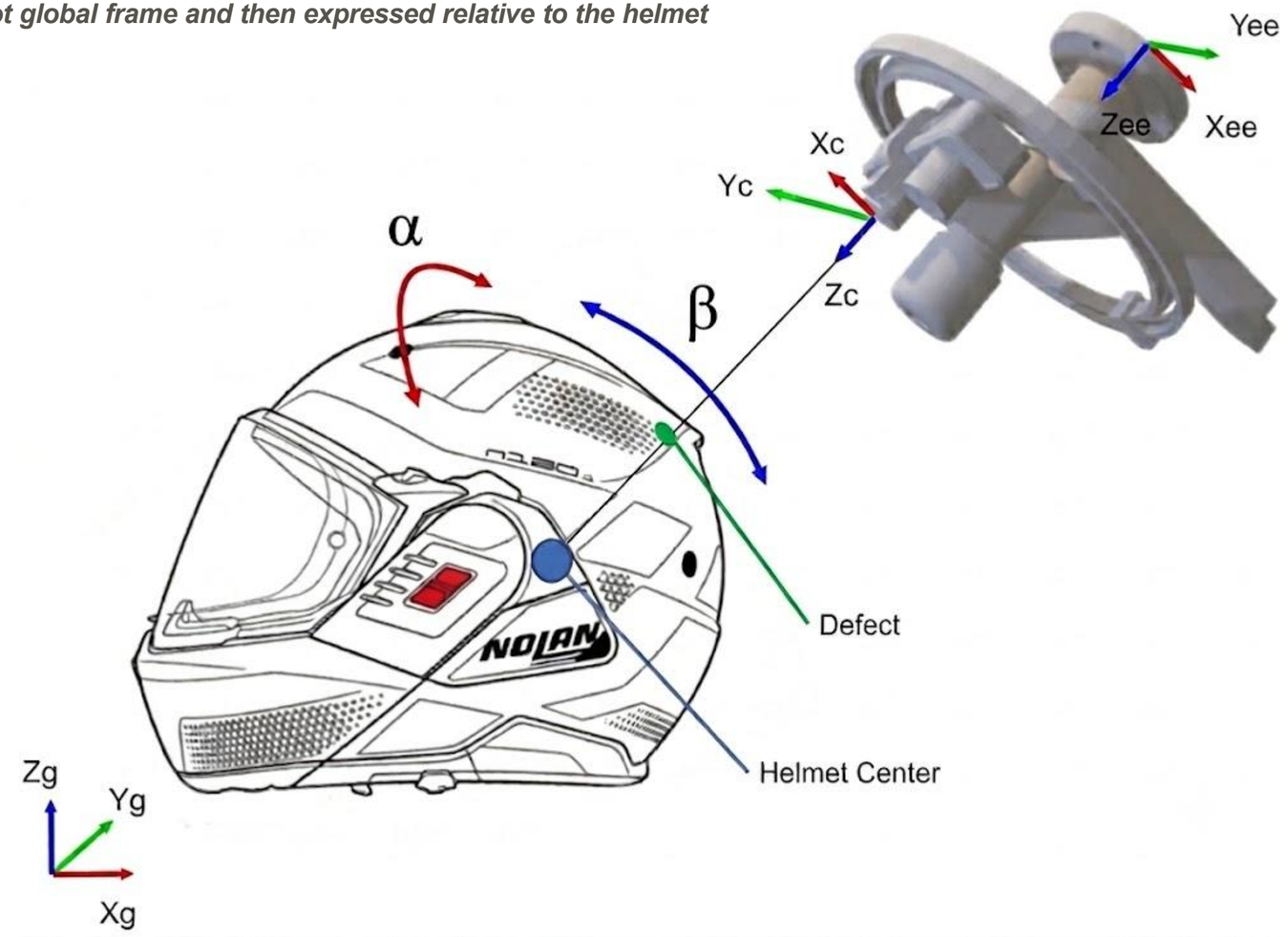
$$H_{(cam \to global)} = H_{(ee \to global)} \cdot H_{(cam \to ee)}$$

02 GLOBAL DEFECT POSITION

$$P_{(global)} = H_{(cam \to global)} \cdot P_{(cam)}$$

03 HELMET-RELATIVE TARGET

[r, α, β] = to_helmet_angles(pos3d_global, helmet_center)



DETECTION

COMPLETE VISION PIPELINE

RGB image and point cloud are converted into helmet-relative defect coordinates

01
Inspection pose
Camera normal to shell

→

02
RGB + point cloud
Zed acquisition

→

03
Colour mask
Green or inverse colours

→

04
2D Filtering
Morphology + Area

→

→

05
3D camera point
Mask pixels → XYZ mean

→

06
Global transform
Camera frame → robot frame

→

07
Spherical transform
Robot frame → Helmet frame

→

08
ALL DEFECT LIST

2D Evidence
Centroid = [cx, cy]
Area
Isolated binary mask
RGB debug image

3D Localization
Points3d
Pos3d_camera = [Xc, Yc, Zc]
Pos3d_global = [Xg, Yg, Zg]
Sph_coord = [r, α, β]

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SECTION 06 / 06

Inspection & Marking

Identifying, refining and marking each detected defect on the shell

01 CONSTRAINING

02 END EFFECTOR

03 HANDLING

04 ACQUISITION

05 DETECTION

06 INSPECTION & MARKING

INSPECTION & MARKING

TOOL POSITIONING

Controlling the tool position via the robot pose

TOOL POSITIONING

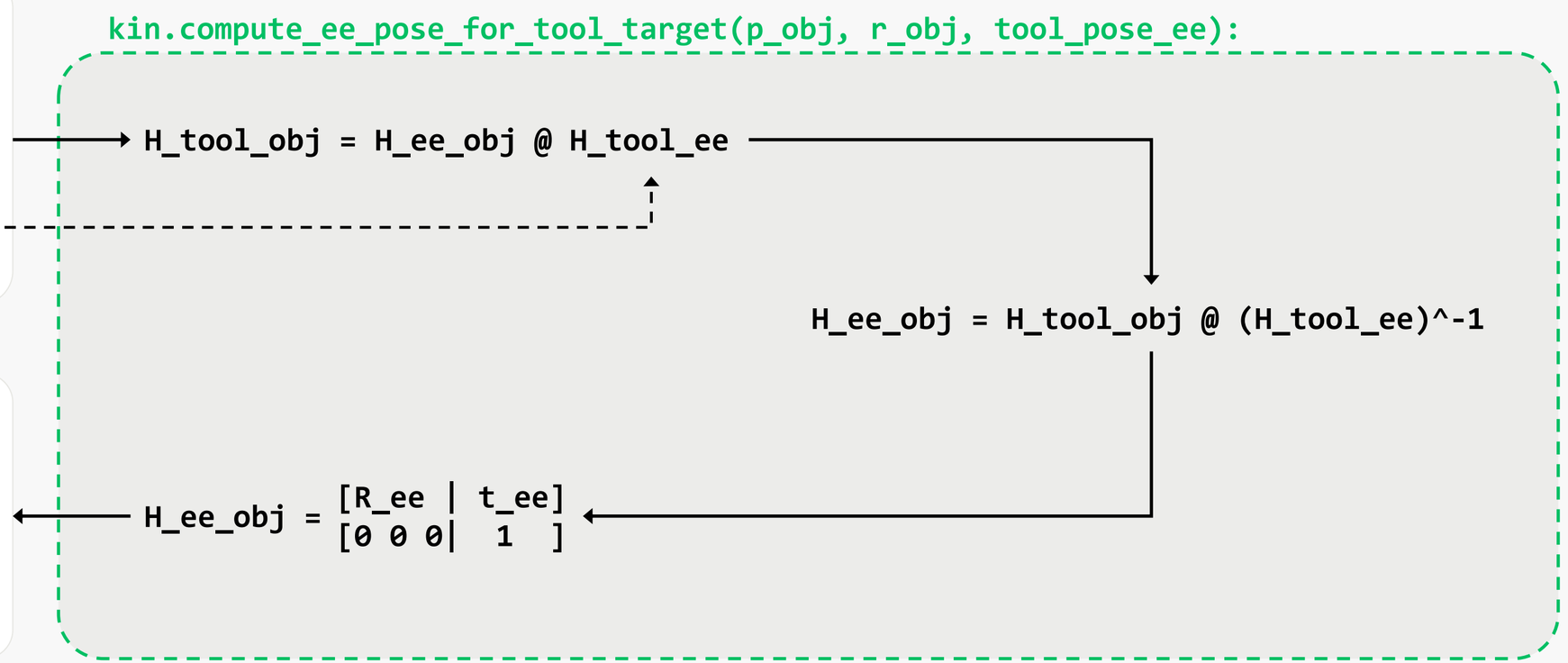
We need to position the tools attached to the end effector

`CAMERA_POSE_EE, MARKER_POSE_EE`

ROBOT CONTROL

All the base movement commands accept the robot's end effector pose

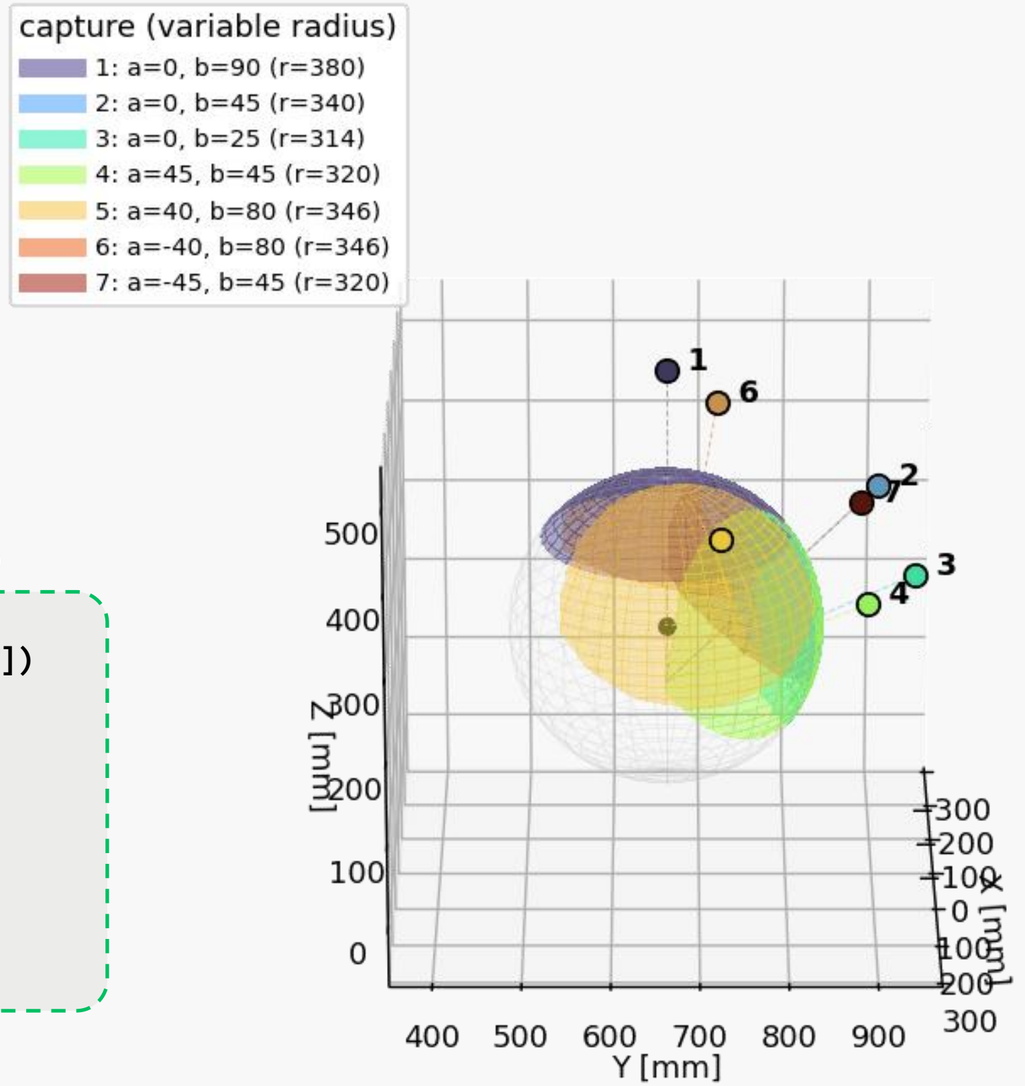
`EE_POSE = [x,y,z,rx,ry,rz]`



INSPECTION & MARKING

HELMET INSPECTION

Initial inspection of the helmet to find all defects



INSPECTION POSITIONS

We manually set multiple inspection positions that will cover all of the helmet's inspectionable surface

```
INSPECTION_POSITIONS = [
    [r, 0, 90],    # top
    [r, 0, 45],    # front1
    [r, 0, 25],    # front2
    [r, 45, 45],   # lat dx1
    [r, 40, 80],   # lat dx2
    [r, -40, 80], # lat sx2
    [r, -45, 45], # lat sx1
]
```

`im.point_and_shoot([r,α,β])`

```
move_circle_spherical([r,α,β])

#image stabilization
time.sleep(1)

take_defects_global()

defect_list.append()
```

INSPECTION & MARKING

HELMET INSPECTION

Filtering of defects identified multiple times

DUPLICATE FILTERING

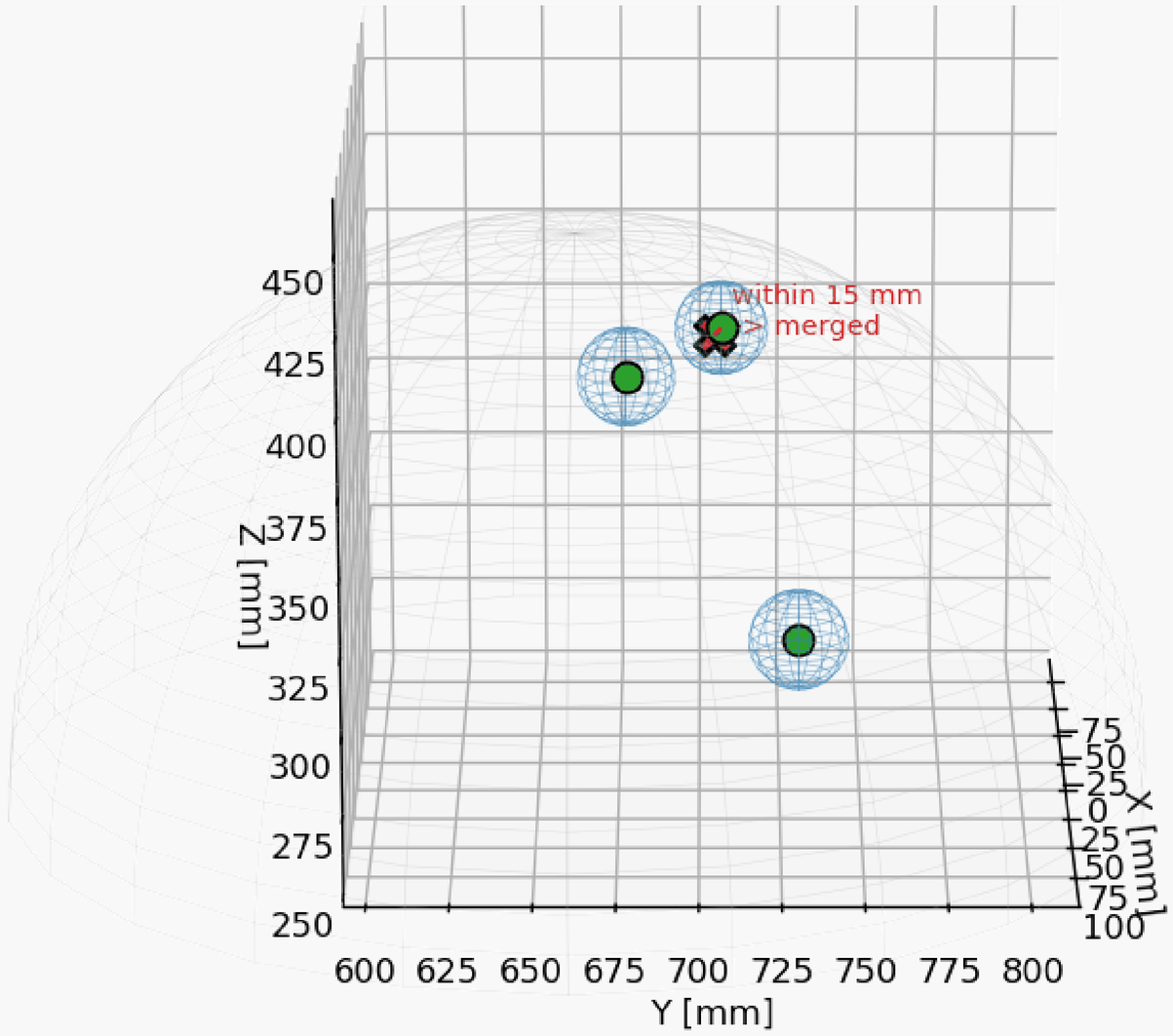
By computing the distance between all defects, we can indetify the ones that are too close (under a certain threshold) and keep only the one with the largest area.

`im.duplicate_filter(defects)`

```
for d in defect_list:

  if dist < threshold:
    keep largest area

  unique.append(d)
```

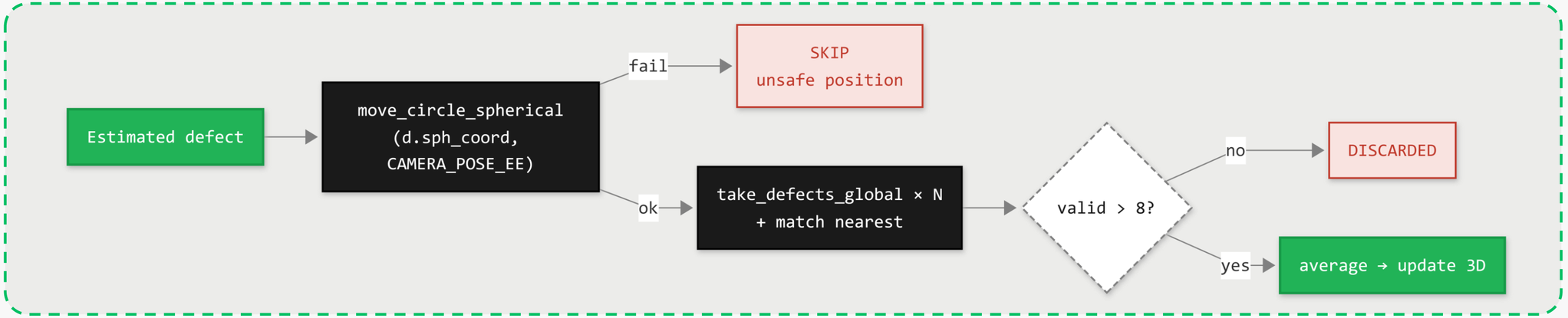


INSPECTION & MARKING

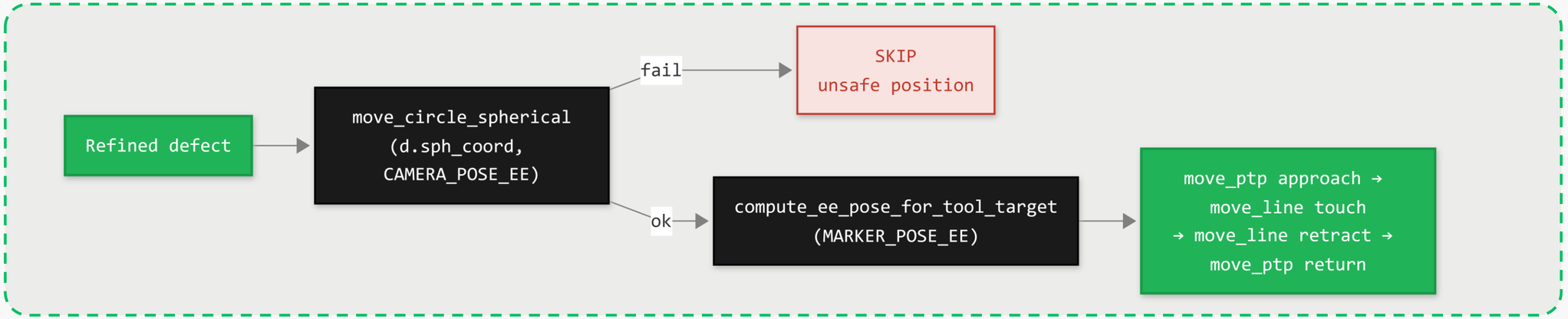
DEFECT REFINEMENT AND MARKING

for defect in unique_defects:

im.refine_defect_position(unique_defect)

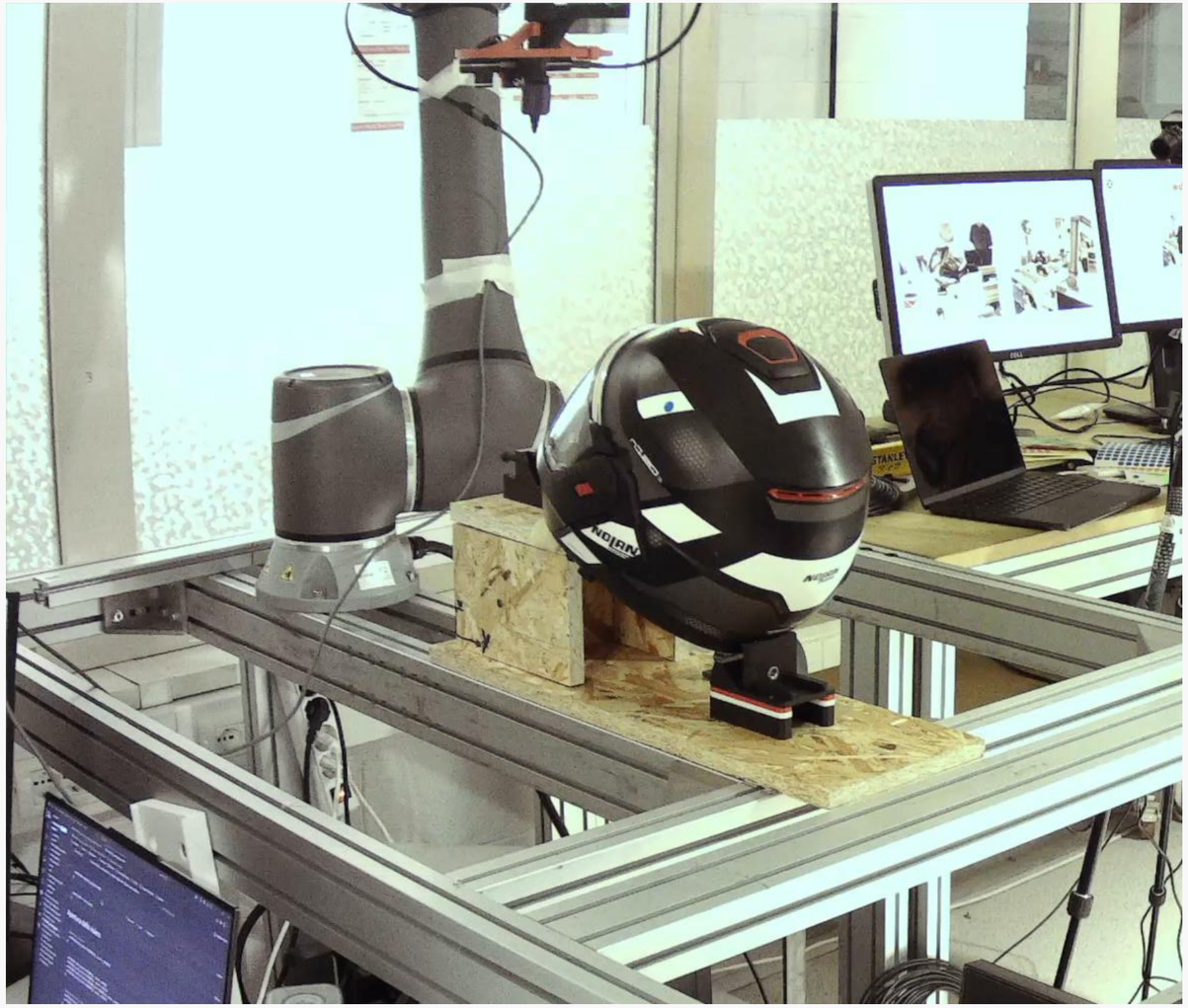


im.mark_defect(refined_defect)



INSPECTION & MARKING

DEFECT REFINEMENT AND MARKING



```
=== GLOBAL INSPECTION ===
Number of inspection positions: 7
```

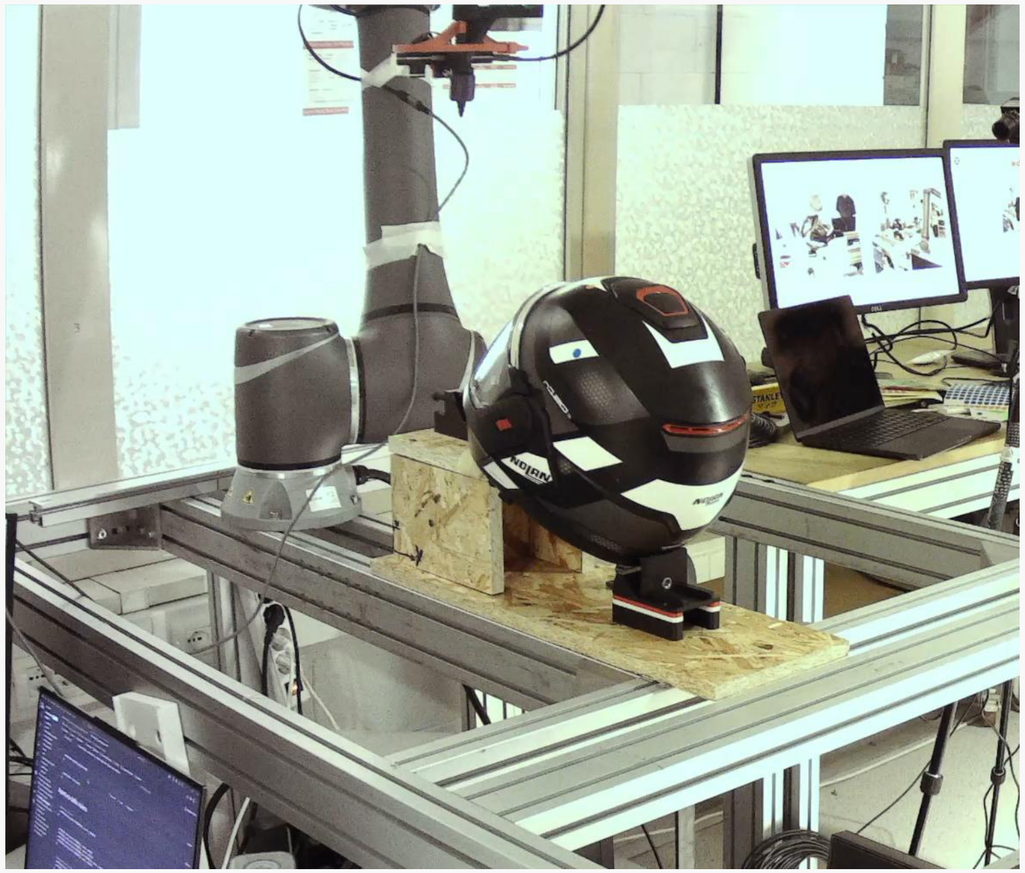


POLITECNICO
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WRAP-UP

CONCLUSIONS

A complete and repeatable inspection cell





01 **EASY AND FAST FIXTURING**

Modular & tool-free, fits both helmet sizes

02 **INTEGRATED END-EFFECTOR**

Handling, vision, lighting & marking in one tool

03 **VISION PIPELINE**

Colored-defects detection located in 3D

04 **AUTOMATIC**

Detect, refine and mark automatically